PRASAD NETHALA

PhD Candidate — Robotics, 3D Perception & Digital Twins | Seeking Summer 2027 Research / Engineering Internship
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SUMMARY

Robotics and 3D perception researcher who takes AI from data capture to deployment on real robots. Built **TomatoPGT**, a published 3D point-cloud dataset and open-source toolset for plant digital twins, and deployed perception and RL models on **NVIDIA Jetson Orin** robots. 3 journal papers, 1 international conference paper, 6 years of prior field engineering.

EDUCATION

Ph.D., Geospatial Computer Science | Texas A&M University–Corpus Christi *Expected Dec **2028***
Dissertation focus: 3D digital twins of plants for agricultural robots (point clouds, scene graphs, GNNs).

M.S., Geospatial Science, Texas A&M Univ.–Corpus Christi (**2023**) | **M.S., Computer Science**, Texas A&M Univ.–Kingsville *2019*

B.Tech., Electrical and Electronics Engineering | Jawaharlal Nehru Technological University, India *2008*

TECHNICAL SKILLS

Languages: Python, C++, RUST
ML / AI: PyTorch, TensorFlow, scikit-learn, Graph Neural Networks, Reinforcement Learning (DDPG / hierarchical RL), 3D segmentation
3D Vision & Perception: Point-cloud processing (Open3D), OpenCV, Structure-from-Motion, photogrammetry, LiDAR–camera calibration, multi-camera rigs
Robotics & Simulation: ROS, NVIDIA Isaac Sim, Gazebo, OpenAI Gym, NVIDIA Jetson Orin, farm-ng Amiga, Fusion 360
Geospatial: ArcGIS Pro, remote sensing, UAV and proximal sensing

RESEARCH EXPERIENCE

Graduate Research & Teaching Assistant (Ph.D.) | Texas A&M University–Corpus Christi *Aug 2023 – Present*

- **TomatoPGT dataset and tools:** built a 3D point-cloud dataset of 42 point clouds tracking three tomato plants across growth stages, annotated with nine custom organ classes (e.g., compound leaf, junction node), and released open-source tools CloudSeg and CloudGraph for organ segmentation and plant-graph construction; published in Data in Brief (2026); code doi:10.5281/zenodo.18651881.
- **Plant scene graphs:** generated organ-level scene graphs (organs and their spatial relations) from point clouds to guide robotic pruning (ASA 2024; AI in Agriculture 2024).
- **Dataset expansion and GNNs (in progress):** scaling TomatoPGT to 12 plants across 3 varieties using video-based 3D reconstruction, and training graph neural networks on plant graphs to predict organ relations and traits.
- **Mobile field photobooth:** converting a farm-ng Amiga robot into a semi-autonomous mobile photobooth that captures plant point clouds in the field for digital twins.
- **Hydroponic harvesting robot:** designed and built a linear-rail harvesting robot end to end (Fusion 360 mechanics, electronics, ROS software) and calibrated its 3-axis arm with a single LiDAR camera (Ubiquitous Robots 2025).
- **Edge deployment:** ran perception and reinforcement-learning models on NVIDIA Jetson Orin onboard robots.
- **Mentoring and teaching:** mentored a capstone team building an autonomous lawn mower; taught ArcGIS Pro and remote-sensing labs.

Research Collaborator | Texas A&M AgriLife Research & Extension Center, Corpus Christi *2024 – Present*

- Applying TomatoPGT methods to AgriLife greenhouse and field phenotyping projects, moving lab digital twins toward production crop research.

Graduate Research Assistant (M.S.) | Texas A&M University–Corpus Christi *Aug 2021 – May 2023*

- Implemented and trained Hierarchical-DDPG, It achieved 38% higher average reward and 2× faster learning than single-agent DDPG and outperformed a hierarchical abstract machines (HAM) baseline (*AI*, MDPI, 2022).

Research Assistant | Machine Learning & Intelligence Lab, University of Louisville *Aug 2020 – May 2021*

- Ran a 42-participant human–robot interaction study (NAO robot) with statistical analysis in Python and R.

PRIOR INDUSTRY EXPERIENCE

Assistant Engineer | Parameshwari Power Projects, Hyderabad, India *May 2009 – Jan 2015*

- Led on-site installation of 30 kV switchyards and 11 kV lines for 2–5 MW power plants (design, budget, crews).

PUBLICATIONS

Nethala, P., et al. "TomatoPGT: A 3D point cloud dataset of tomato plants for segmentation and plant-trait extraction." Data in Brief, 112642 (2026).

Nethala, P., et al. "Techniques for Canopy to Organ Level Plant Feature Extraction via Remote and Proximal Sensing: A Survey and Experiments." Remote Sensing, MDPI (**2024**, *16*(23), 4370).

Um, D., **Nethala, P.,** & Shin, H**. (2022).** Hierarchical DDPG for manipulator motion planning in dynamic environments**.** AI, 3(3), 645–658. doi:10.3390/ai3030037.

Nethala, P. "Calibration of a Three-Axis Robot Manipulator with a Single LiDAR Camera for Indoor Hydroponic System." International Conference on Ubiquitous Robots (UR),2025.